

difes against FOCUS

A validation on Irati Tolosa's chromium data: the same spectra, the same models, the same path

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1 Purpose

The TFG of Irati Tolosa (2025) measured the spectral emissivity of pure and oxidised chromium between 200 °C and 700 °C and fitted the 500 °C reflectances with FOCUS 2.0 (CEMHTI, Orléans): the pure metal with ε_∞ and two Drude terms, the oxidised sample as a Cr_2O_3 film on that metal. The group's own tool, **difes** (dielectric function, inversion, fitting, emissivity, simulation), was built to do the same work on macOS and Linux. This document answers two questions with numbers:

1. **Does difes compute what FOCUS computes?** Same formulas, same parameters, same curves, to the precision FOCUS writes.
2. **Does difes, used as Irati used FOCUS, land on her results?** Following her path from the bare metal to the final anisotropic film, step by step.

Everything below comes from two scripts in the difes repository, `examples/reproduce_irati.py` and `examples/irati_path.py`, run on 1 and 2 October 2026 with difes 0.14.1. The inputs are Irati’s measured spectra at 10° incidence, the two project files she exported from FOCUS (`Bestebat.sml`, the intermediate project; `Azkena.sml`, the final one), three curves FOCUS itself wrote out of `Azkena.sml`, and the curve of figure 20 of the report (`curve.dat`).

2 What “reproduce” can mean

A fit program gives three things: the curve it draws for a model, the parameters it stops at, and the way a person reaches them. These are not equally reproducible, and the report itself says so about the second: the second Drude term of chromium is called unreliable, and the oxide fit “does not reach the 550 cm⁻¹ peak” and “fails at the edges”. The test is therefore layered.

- **Level A, forward optics.** Same parameters must give the same curve. This must be exact.
- **Levels B and C, the fit as a minimisation.** From the report’s values and from elsewhere, does difes’s fitter find the same curve, and the same parameters where the problem is well posed?
- **Level D, the procedure.** The guided fit of difes’s Bench on the emissivity itself, as a student would do it.
- **Irati’s path.** The five models of the work rebuilt in order and compared, at each step, with what FOCUS gave.

3 Level A: forward optics is exact

3.1 FOCUS’s own output against difes

Irati exported three curves of `Azkena.sml` from FOCUS (512 points on 0–1500 cm⁻¹, seven significant digits). difes reads the same project file, turns every definition into a formula of its math language, and evaluates the same definitions on FOCUS’s grid.

Table 1: FOCUS’s exported curves against difes’s evaluation of the same definitions.

FOCUS export (file)	the same definition in difes	largest difference
<code>Azkena-PhysRev_doitzeko</code>	<code>R_PhysRev_doitzeko</code> , the final reflectance R	5.6×10^{-6} , rms 8.6×10^{-7}
<code>Azkena-teoriko_monokristal</code> and <code>Azkena-oxido</code> (identical files)	<code>Funtzio_dielektrikoa_PhysRev</code> , the ε of the single-crystal model	4.4×10^{-5} relative

Both differences are the seven-digit rounding of FOCUS’s own files (the reflectance to 5×10^{-7} ; ε reaches 128 in the band). Number against number, nothing transcribed, nothing guessed: where FOCUS wrote its result down, difes reproduces it to the last digit written. This comparison is a test of the package and runs with its test suite.

3.2 FOCUS’s formulas against difes’s own optics

The project files also hold FOCUS’s formulas for the plate optics. At identical parameters, difes’s native sample optics give:

Table 2: difes’s sample optics against FOCUS’s formulas at the same parameters.

quantity (Bestebat.sml, 400–1350 cm ⁻¹)	difes form	rms difference
bare chromium, Fresnel reflectance	—	1.5×10^{-16}
0.8 μm incoherent oxide film	focus (eq. 27 of the report)	1.7×10^{-16}
0.8 μm incoherent oxide film	energy (difes’s default)	8.8×10^{-3} , at most 0.049

The third row is the one finding of this level. FOCUS’s incoherent film, its **Reflectance-Ei-E1-E2** symbol and eq. 27 of the report, writes the numerator as $|(1 - r_1^2) r_2 \tau^2|^2$, the product of amplitude transmittances; difes’s default writes $(1 - R_1)^2 R_2 A$, the energy-conserving intensity form. They coincide when the film’s index is real and differ where the film absorbs, up to 0.05 in the reststrahlen band of Cr₂O₃. difes offers both since 0.14.0, so a FOCUS result can be reproduced exactly and the physical form remains available.

3.3 The report’s model against its figure

A third check uses the report rather than the files: Table 4 and Table 6 (fourteen Lorentz oscillators; ω_0 of the fourth is not legible in the transcription) on an incoherent film of unstated thickness, against the curve of figure 20. Fitting only those two unknowns:

Table 3: The report’s tables against the curve of its figure 20.

film form	ω_0 of oscillator 4	thickness	rms vs curve.dat	above 450 cm ⁻¹
energy	625.0 cm ⁻¹	1.093 μm	9.9×10^{-3}	3.7×10^{-3}
focus	624.7 cm ⁻¹	1.196 μm	1.8×10^{-2}	1.6×10^{-2}

The curve of the report is reproduced to the precision of its transcribed parameters. Which film symbol drew that figure cannot be decided from the tables; the files, as the path below shows, point to the intermediate project.

4 Level B: pure chromium

The report fits ε_∞ plus two Drude terms to $R = 1 - E$ of the pure sample at 500 °C and 10°, excluding the 1200–2400 cm⁻¹ dip it attributes to the measurement. On the same 545 points:

Table 4: difes’s fits of the pure metal under two weightings and three starting points.

weights	start	rms vs data	rms vs Table 4 curve	ε_∞	$(\omega_p, \gamma)_1$	$(\omega_p, \gamma)_2$
Table 4 itself		0.0075	—	4.86	(15344, 834)	(26930, 14258)
measured σ	Table 4 · Table 4 × 0.7 · generic	0.0069	0.0037	42.1	(30211, 2485)	(10 ⁵ , 77796)
uniform	Table 4 · Table 4 × 0.7 · generic	0.0071	0.0017	1.00	(13043, 689)	(51020, 10 ⁵)
uniform, ε_∞ frozen at 4.86	generic	0.0071	—	4.86	(17629, 1046)	(82313, 10 ⁵)

Three starting points give the same answer under each weighting, so the fitter converges. Its curves lie within 0.002–0.004 of the Table 4 curve, less than the scatter of the data (median σ 0.0015, Table 4’s own residual 0.0075), and its residual is lower than Table 4’s. The parameters, however, are not Table 4’s and depend on the weighting, with the second Drude term running to its bounds. A metal’s reflectance

near 0.95 fixes the combination of ε_∞ and the two plasma frequencies, not each of them: the problem is degenerate, as the report’s own remark about the second oscillator says. Reproducing Table 4’s numbers would require the unrecorded choices of where the fit was stopped. What difes reproduces is the physical content, the curve, from any start.

5 Level C: the oxide film

Table 6 on a frozen Table 4 substrate, incoherent film of 1.095 μm with the FOCUS form, fitted over 390–1300 cm^{-1} as the report says (118 points):

Table 5: difes’s refits of the oxide film from the report’s values and from a perturbed start.

weights	start	rms vs data	rms vs curve.dat	ε_∞	ω_0 shifts rms / max	within their γ
Table 6 as transcribed		0.074	0.026	1.87	—	—
FOCUS’s own curve (curve.dat)		0.046	—			
measured σ	Table 6	0.035	0.028	4.0	24 / 60 cm^{-1}	14 of 14
measured σ	Table 6, perturbed	0.032	0.028	3.3	27 / 60 cm^{-1}	13 of 14
uniform	Table 6	0.031	0.027	3.4	22 / 44 cm^{-1}	13 of 14
uniform	Table 6, perturbed	0.031	0.027	3.4	22 / 44 cm^{-1}	13 of 14

FOCUS’s own fitted curve sits 0.046 from the data over the range, and Table 6 as transcribed 0.074. difes’s refits reach 0.031–0.035, from the report’s values and from a perturbed start alike, and keep thirteen or fourteen of the fourteen oscillators within their own width of the report’s positions. The ones that move are the ones the report flags: the first mode drifts to its bound at 400 cm^{-1} (the edge the report could not fit), the broad thirteenth at 1127 cm^{-1} with $\gamma = 495 \text{ cm}^{-1}$ is replaced by a narrower one, and the last moves to the upper edge. The agreement is where the physics is, in the bands; the disagreement is where the report says its own fit was weak.

6 Level D: the Bench’s way

difes’s simple skin, the Bench, runs a guided fit in four stages: strengths (and ε_∞), then positions, then widths, then everything. On the emissivity itself, with the film sample and the FOCUS formula, over 390–1300 cm^{-1} :

Table 6: The guided fit of the Bench from the report’s values and from its own suggestions.

start	strengths $\rightarrow$ positions $\rightarrow$ widths $\rightarrow$ all (rms in E)	adopted	bands nearest Table 6’s 437, 493, 577, 624 cm^{-1}
Table 6	0.062 $\rightarrow$ 0.032 $\rightarrow$ 0.032 $\rightarrow$ 0.032	14 terms, ε_∞ 3.3; its R is 0.028 from curve.dat	15, 41, 1, 1 cm^{-1} away
the Bench’s suggestions	0.111 $\rightarrow$ 0.092 $\rightarrow$ 0.056 $\rightarrow$ 0.056	6 terms, ε_∞ 4.9; its R is 0.037 from curve.dat	15, 21, 63, 111 cm^{-1} away

From the report’s values the Bench gives the fitter’s result in four visible stages. From its own suggestions, found in the fit range, it recovers the two strongest bands but not the 577 and 624 cm^{-1} pair or the weak overtones above 700 cm^{-1} , which the report needed Larbi’s assignments and a hand-built list of fourteen terms to place. A model of that kind is prior knowledge, and importing the FOCUS model is how it enters.

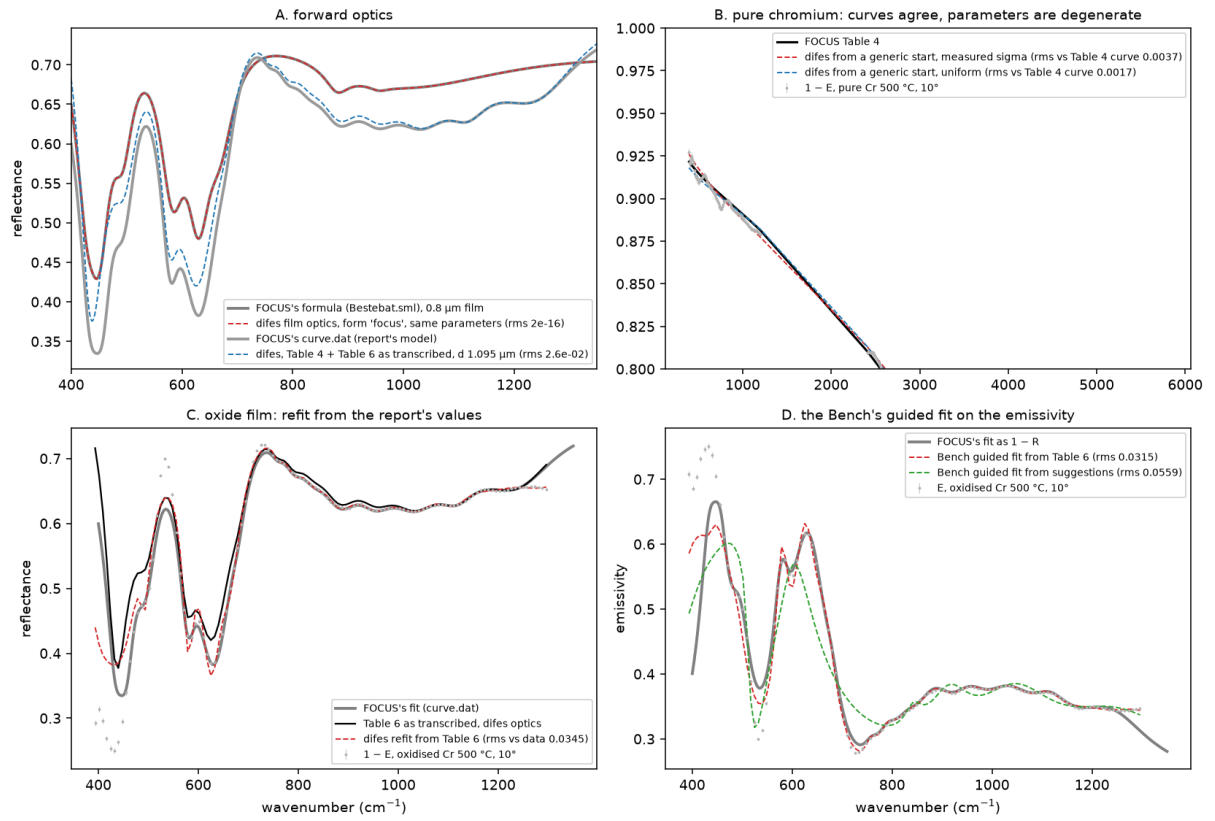


Figure 1: Levels A to D. A: FOCUS's film formula from Bestebat.sml and difes's film optics with the FOCUS form coincide; the report's model against FOCUS's drawn curve. B: pure chromium, the Table 4 curve and difes's fits from a generic start under two weightings. C: the oxide film, Table 6 as transcribed and difes's refit from it. D: the Bench's guided fit on the emissivity from Table 6 and from the Bench's own suggestions.

7 Irati's path, step by step

The five models of the work, rebuilt in order as difes sessions and compared at each step with what FOCUS gave. Each is also an example session of the app (“Irati's path · 1” to “5”).

Table 7: The five steps of Irati's path.

step	Irati's model	in difes	against FOCUS	vs measurement
1 · the metal	ε_∞ 4.859, Drude (15344, 834) and (26930, 14258): Table 4, and the Funtzio_dielektrikoa of both files	a model node and a bulk sample	difes's Fresnel reflectance and FOCUS's Reflektibitatea formula: 10^{-16}	rms 0.010 on 400–6000 cm^{-1}
2 · the oxide as a film	ε_∞ 6.1 and eight phonons (Bestebat), 0.8 μm incoherent film with eq. 27	a model node and a film sample with the FOCUS form	difes's sample optics and FOCUS's Reflektibitatea_oxido : 10^{-16} ; against the report's curve 0.055	0.078
3 · the effective layer	a 0.5 μm Bruggeman mixture of air and oxide at 50 % over the film (Bestebat)	FOCUS's own formulas as expression nodes	against the report's curve 0.022: this is the model of figure 20	0.041
4 · the literature model	the single-crystal Cr_2O_3 dielectric function (Azkena)	the definition evaluated on FOCUS's grid	FOCUS's exported ε : 4×10^{-5} relative, the file's rounding	—
5 · the final model	$\frac{1}{3}\varepsilon_{\parallel} + \frac{2}{3}\varepsilon_{\perp}$, 0.7 μm film under a 0.3 μm Bruggeman layer (Azkena)	the definition evaluated on FOCUS's grid	FOCUS's exported R : 6×10^{-6} at most, the file's rounding	0.124

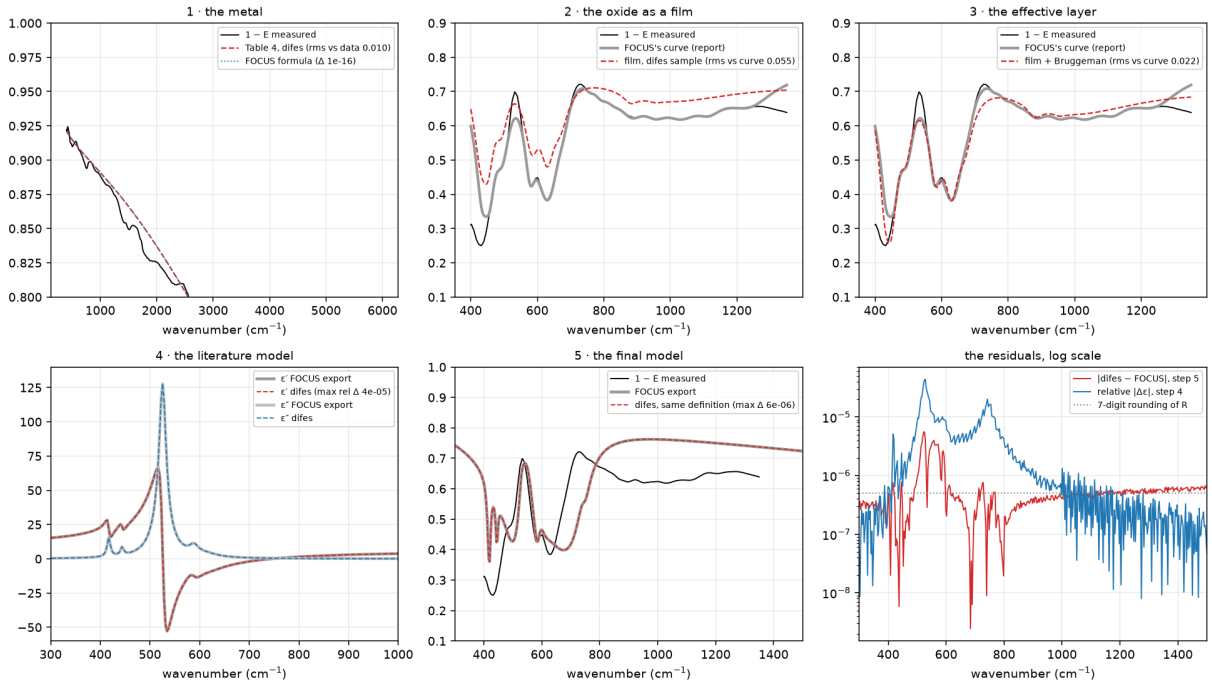


Figure 2: Irati's path. 1 the metal; 2 the oxide as a 0.8 μm film; 3 the film under a Bruggeman layer; 4 the single-crystal model against FOCUS's export; 5 the final anisotropic model against FOCUS's export; and the residuals of 4 and 5 on a log scale, with the seven-digit rounding of FOCUS's files as the dotted line.

Three things follow. **Same numbers:** where FOCUS wrote its output down, difes reproduces it to the

last digit written; where only formulas exist, difes’s native optics and FOCUS’s formulas agree to machine precision at the same parameters once the film formula is FOCUS’s. **Same path:** the substrate of step 1 is the same in the report and in both files; the curve of the report’s figure 20 is step 3, the film under the Bruggeman layer, and the final file records an attempt that did not improve on it (0.124 against the measurement where step 3 gives 0.041). **What is open:** the Bruggeman layer runs in difes as FOCUS’s formula; making it a native sample geometry, so that step 3 can be refitted and not only reproduced, is the next item of the roadmap.

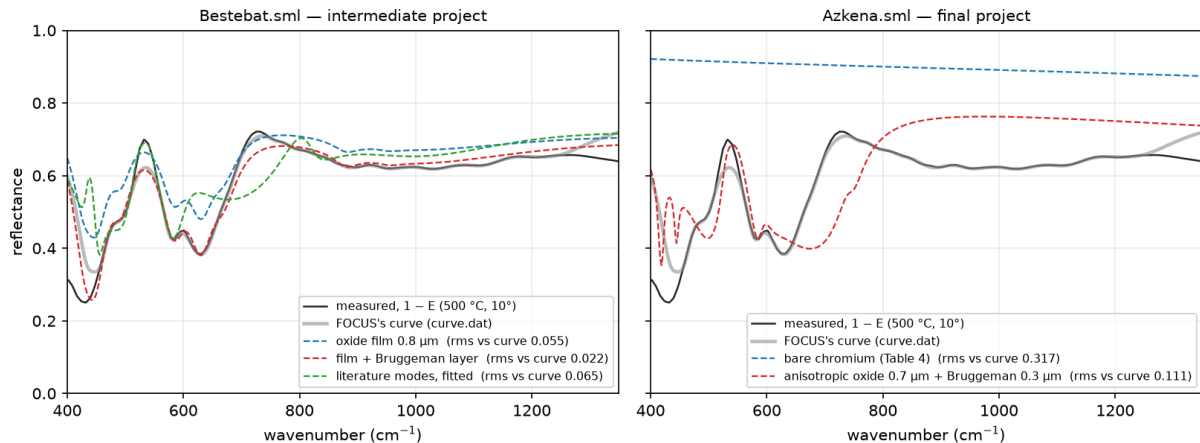


Figure 3: The two FOCUS projects evaluated by difes on FOCUS’s grid (dashed), FOCUS’s curve of the report (grey) and the measured reflectance $1 - E$ at 500 °C (black). The intermediate project’s film-plus-Bruggeman reflectance is the curve of the report; the final project’s definition is not.

8 Verdict

- **Exact where it must be.** With the same formula, difes computes FOCUS’s curves to the precision FOCUS writes; the formula FOCUS uses for an incoherent film is an option of difes, and its difference with the energy-conserving form is documented.
- **Convergent and consistent.** The fitter reaches the same solution from the report’s values and from generic starts, with lower residuals than the tabulated solutions; its curves agree with FOCUS’s within the data’s scatter.
- **Parameters agree where the problem is well posed.** The oxide’s band positions, within their widths; chromium’s two Drude terms are degenerate for either program, as the report acknowledges.
- **Same place on Irati’s path.** Every step of the work is reproduced; the figure of the report is identified with the intermediate project’s model.

9 Appendix: files, versions, how to rerun

Measured spectra. `cr_pure_500C_E_10deg.txt` and `cr_oxidised_500C_E_10deg.txt`: the 10° columns of Irati’s emissivity files at 500 °C, converted to $R = 1 - E$ where the report does so.

FOCUS files. `Bestebat.sml` (17 symbols and definitions) and `Azkena.sml` (14), exported from FOCUS 2.0 with embedded declarations; `Azkena-PhysRev_doitzeko.dat`, `Azkena-oxido.dat`, `Azkena-teoriko_monokristal.dat`, FOCUS’s own output curves; `curve.dat`, the fitted reflectance of figure 20 of the report.

Software. difes 0.14.1 (Python 3.12, NumPy, SciPy), its shell 0.8.2 for macOS. The validation runs as tests of the package (`tests/test_osml.py`) and as two example scripts:

```
.venv/bin/python examples/reproduce_irati.py
.venv/bin/python examples/irati_path.py
```

Where to read more. The project site holds the manual (using difes; theoretical basis), the log of every step, the release notes, and the two pages this document condenses, “Validation against FOCUS” and “Irati’s path, step by step”.